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TRW Automotive

Slip Control Software Design

Document ID	:	ARM-DDD-Main
Title	:	DDD for ARM Subsystem
Summary	:	Description of subsystem architecture and interface
Scope	:	Application: • arm{.c, .h} • arm_ay_mod{.c, _priv.h, _sub.h} • arm_calibration_database_sub.h • arm_input{.c, _sub.h} • arm_logger.h • arm_output{.c, _sub.h} • arm_roll_over_index{.c, _priv.h, _sub.h} • arm_state{.c, _sub.h} • arm_trims{.c, .h, _sub.h} • arm_utilities{.c, _sub.h} • arm_veh_control{.c, _priv.h, _sub.h} • arm_veh_state{.c, _priv.h, _sub.h} • arm_version.h • arm_whl_control{.c, _priv.h, _sub.h} • ay_limiter{.c, _priv.h, _sub.h, _trim.h, _trim_sub.h} • ay_limiter_output_priv.h Calibration: • arm_calibration_database{.c, .h} • calibration_arm{.c, .h, _priv.h} • calibration_ay_limiter{.c, .h, _priv.h}
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REFERENCES

Ref.	Document	Title	Ver.	Location
1	ARM-DDD-Trimmanual.pdf	ARM-DDD-Trimmanual	1.0	ARM-DDD-Trimmanual

REVISION HISTORY

Rev.	Date dd/mm/yyyy	Author	Change Request	Comments (Version)
1.0	30/05/2011	B. Etges		Initial Version
1.1	11/01/2013	B. Etges		Add IsDynArmActive()
1.2	23/08/2016	A. Hoehn		Update to AR 16

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1 ABSTRACT

The Active Roll Mitigation ([ARM](#)) function is designed to prevent vehicle rollover by brake and engine intervention. Dynamic roll over as well as steady state roll over are considered. [ARM](#) relies on early and fast pressure commands to reduce energy in critical situations.

2 ACRONYMS

ARM Active Roll Mitigation

AyLimiter Control Algorithm that limit the Lateral Acceleration of the Vehcile

COG Center of Gravity

RSC Roll Stability Control

ESC Electronic Stability Control

YSC Yaw Stability Control

WSR Wheel Slip Regulator

3 OVERVIEW

This document describes

- all inputs that are required by Arm
- all outputs provided by Arm
- all tasks provided internally or to the operating system in order to schedule the sub-system

3.1 DESIGN GOAL

Main goal of [ARM](#) function is to prevent vehicle rollover. Dynamic roll over as well as steady state roll over are considered.

[ARM](#) algorithm evaluates the vehicle conditions, determines whether an impending rollover condition exists, and determines whether to intervene. When [ARM](#) activates, it provides pressure and slip commands to the appropriate wheels in an attempt to prevent rollover by reducing the vehicle speed and energy.

3.2 GENERAL REQUIREMENTS

1. ARM should prevent dynamic rollover.
2. ARM should prevent static rollover.

3.3 DETAILED REQUIREMENTS

Requirement ID	
Location	

3.4 SYSTEM MODE REQUIREMENTS

Depending on the vehicle configuration (i.e. loading, chassis level) and driver switches (of-froad, roof load) ARM has to be sensitized or desensitized.

3.5 FLOW-CHARTS AND DIAGRAMS

This flowchart, Fig. 3.1, shows the high level signal flow and system architecture environment of ARM in the Electronic Stability Control (ESC).

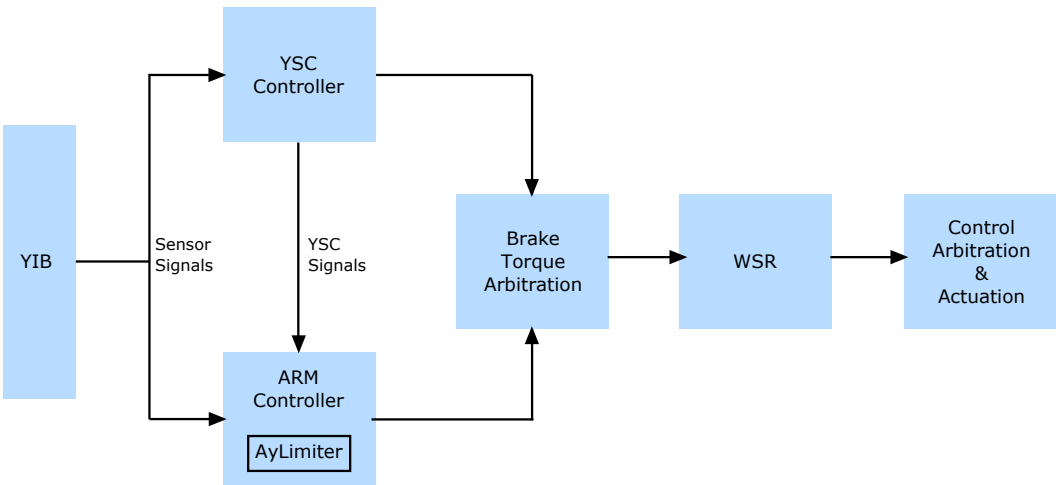


Figure 3.1: System Architecture Environment of ARM in ESC

The following flow chart, Fig. 3.2, shows the high level signal flow within ARM.

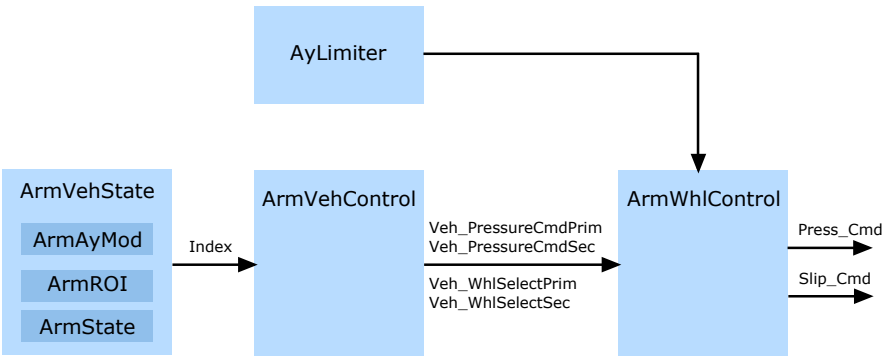


Figure 3.2: System Architecture Environment of ARM in ESC

4 INTERRUPTS

[ARM](#) does not use any interrupt operations.

5 REQUIRED INPUTS

Reference interface descriptions in chapter 6.

All input signals for the subsystem [ARM](#) are described in section 6.1 InitArm and section 6.3 UpdateArmInput. An brief overview of all input variables is available in Fig. 5.1.

ARM_INPUT_TYPE_TAG
+ Abs_Ay_Filt_Max: S16
+ Arm_Brake_Enabled: BOOLEAN
+ Arm_Enabled: BOOLEAN
+ Arm_Engine_Enabled: BOOLEAN
+ Ay_Ov_U: S16
+ Ay_Uncert_Filt_State: T1POLESTATESCANONIC
+ Brake_Pressure: S16 [WHEEL_MAX]
+ Characteristic_Velocity: S16
+ Chassis_Level_State: U8
+ Drive_Torque: S16
+ Driver_Brake_Pressure_Req_Val: BOOLEAN
+ Driver_Brake_Pressure_Request: S16
+ Driver_Mode_Switch_Offroad: BOOLEAN
+ Driver_Requested_Torque: S16
+ Lat_Accel: S16
+ Lat_Accel_Deriv: S16
+ Lat_Accel_Deriv_Slow: S16
+ Lat_Accel_Deriv_Slow_State: VARFILTSTATES
+ Lat_Accel_Uncertainty: S16
+ Mue_Estim_s16: S16
+ R_Lin: S16
+ R_Lin_Deriv: S16
+ R_Lin_Deriv_States: FILTER_DERIVATIVE_TYPE
+ R_Ref: S16
+ Reverse_Detected: BOOLEAN
+ Road_Wheel_Angle: S16
+ Road_Wheel_Angle_Deriv: S16
+ Roof_Load_State: U8
+ System_Driver_Decel_Pressure: S16 [WHEEL_MAX]
+ Veh_Velocity: S16
+ Vehicle_Load_State: U8
+ Wheel_Base: S16
+ Yaw_Rate: S16
+ Yaw_Rate_Deriv: S16
+ Ysc_Forced_Understeer: BOOLEAN

Figure 5.1: Input variables

6 PROVIDED TASKS

6.1 InitArm

Description	InitArm
Rate	Once at startup of μC
Dependency	
SYNTAX	void InitArm (void)

6.2 ReInitArm

Description	ReInitArm
Rate	Once when μC is reinitialized
Dependency	
SYNTAX	void ReInitArm (void)

6.3 UpdateArmInput

Description	UpdateArmInput
Rate	Not slower than 10 msec
Dependency	UpdateArmInput has to be scheduled before UpdateArmProcessing
SYNTAX	void UpdateArmInput (void)

6.4 UpdateArmProcessing

Description	UpdateArmProcessing
Rate	Not slower than 10 msec
Dependency	UpdateArmProcessing has to be scheduled after UpdateArmInput and before UpdateArmOutput
SYNTAX	void UpdateArmProcessing (void)

6.5 UpdateArmOutput

Description	UpdateArmOutput
Rate	Not slower than 10 msec
Dependency	UpdateArmOutput has to be scheduled after UpdateArmProcessing and before UpdateYscMidTaskInput
SYNTAX	void UpdateArmOutput (void)

7 PROVIDED SERVICES

ARM provides several services for subsystem communication. An overview is presented in Fig. 7.1.

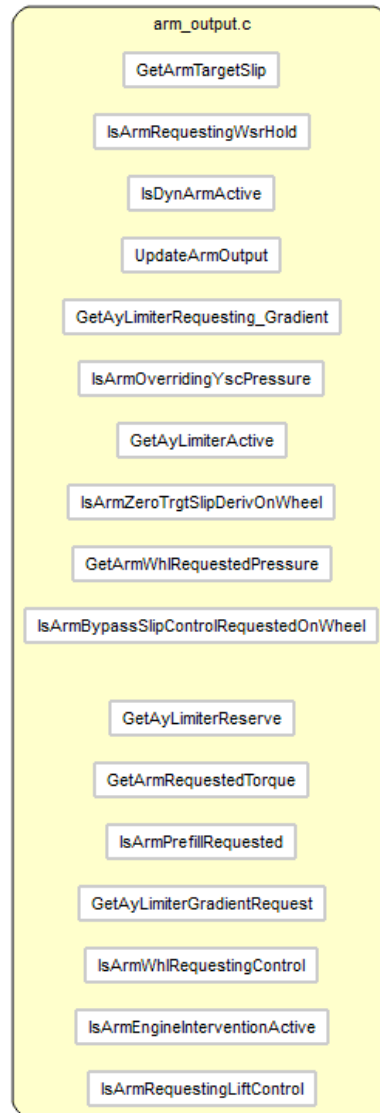


Figure 7.1: Input variables

7.1 GetArmTargetSlip

GetArmTargetSlip

7.2 GetArmWhlRequestedPressure

GetArmWhlRequestedPressure

7.3 IsArmWhlRequestingControl

IsArmWhlRequestingControl

7.4 IsArmOverridingYscPressure

IsArmOverridingYscPressure

7.5 IsArmPrefillRequested

IsArmPrefillRequested

7.6 IsDynArmActive

IsDynArmActive

7.7 GetAyLimiterActive

GetAyLimiterActive

7.8 GetAyLimiterReserve

GetAyLimiterReserve

7.9 GetAyLimiterGradientRequest

GetAyLimiterGradientRequest

7.10 GetAyLimiterRequesting_Gradient

GetAyLimiterRequesting_Gradient

7.11 GetArmRequestedTorque

GetArmRequestedTorque

7.12 IsArmEngineInterventionActive

IsArmEngineInterventionActive

7.13 IsArmRequestingWsrHold

IsArmRequestingWsrHold

7.14 IsArmBypassSlipControlRequestedOnWheel

IsArmBypassSlipControlRequestedOnWheel

7.15 IsArmRequestingLiftControl

IsArmRequestingLiftControl

7.16 IsArmZeroTrgtSlipDerivOnWheel

IsArmZeroTrgtSlipDerivOnWheel

8 CONFIGURATION OPTIONS

Name	Type	Range	Description
AY_LIMITER	Boolean	[0 ..1]	En-/Disable AyLimiter functionality
AYLIM_ARM_CTRL_SAME_TIME	Boolean	[0 ..1]	En-/Disable Arm and Aylimiter to be active and in control at the same time. In case of disabled configuration, Arm has higher priority than AyLimiter
ENABLE_ARM_REAR_WHL	Boolean	[0 ..1]	En-/Disble Arm and Aylimiter control at the rear axel.

9 CONFIGURATION PROCEDURES

Depending on project requirements and the roll over propensity of the vehicle the configurations has to be adjusted once before start of project. Changes during development phase of the project can be applied but should be carefully reviewed with the subsystem responsible Team lead.

10 TEST CASES

Test category	
Requirements	
Test Case ID	

11 APPENDIX

12 CALIBRATION INFORMATION

The calibration procedure is outlined in ARM-DDD-Trimmanual, see References for further information.